

Solution Problem 1.3

$$(a) \quad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & & \\ \vdots & & & \\ \sigma_{p1} & & & \sigma_{pp} \end{bmatrix} \quad V = \begin{bmatrix} \sigma_{11} & & 0 \\ & \sigma_{22} & \\ 0 & & \ddots \\ & & & \sigma_{pp} \end{bmatrix}$$

$$\Sigma V^{-1/2} = \begin{bmatrix} \sigma_{11}/\sqrt{\sigma_{11}} & \sigma_{12}/\sqrt{\sigma_{22}} & \dots \\ \sigma_{12}/\sqrt{\sigma_{11}} & \sigma_{22}/\sqrt{\sigma_{22}} & \\ \vdots & & \end{bmatrix}$$

$$V^{-1/2} \Sigma V^{-1/2} = \begin{bmatrix} 1 & \frac{\sigma_{12}}{\sqrt{\sigma_{11}\sigma_{22}}} & \dots \\ & 1 & \\ & & \ddots \\ & & & 1 \end{bmatrix} = S \quad \text{QED}$$

$$S = V^{-1/2} \Sigma V^{-1/2} \Leftrightarrow V^{1/2} S = \Sigma V^{-1/2} \\ \Leftrightarrow V^{1/2} S V^{1/2} = \Sigma \quad \text{QED}$$

$$(b) \quad \Sigma = V^{1/2} S V^{1/2}$$

$$|\Sigma| = |V^{1/2}| |S| |V^{1/2}|$$

$$\text{Since } |V^{1/2}| = \sqrt{\sigma_{11}\sigma_{22}\dots\sigma_{pp}}$$

$$|\Sigma| = \sigma_{11}\sigma_{22}\dots\sigma_{pp} |S|$$

QED

$$(c) \quad \Sigma = P \Lambda P^T \quad (\text{eigenvalue decomposition}) \quad \Sigma \vec{e}_i = \lambda_i \vec{e}_i, i=1, \dots, p$$

$$\text{where } P = [\vec{e}_1 \dots \vec{e}_p], \quad \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix}$$

$$\text{and } P P^T = I \Leftrightarrow P^{-1} = P^T \quad (\text{orthogonal}) \\ \Leftrightarrow |P| = 1$$

$$|\Sigma| = |P| |\Lambda| |P^T| = 1 \cdot \lambda_1 \lambda_2 \dots \lambda_p \cdot 1 = \prod_{i=1}^p \lambda_i \quad \text{QED}$$

$$(d) \quad \Sigma = \begin{bmatrix} \sigma_{11} & g\sqrt{\sigma_{11}\sigma_{22}} \\ g\sqrt{\sigma_{11}\sigma_{22}} & \sigma_{22} \end{bmatrix}$$

$$|\Sigma| = (1 - g^2) \sigma_{11} \sigma_{22}$$

$$\min_g |\Sigma| = 0 \quad \text{for } |g| = 1$$

$$\max_g |\Sigma| = \sigma_{11} \sigma_{22} \quad \text{for } g = 0$$

$$c) \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Sigma e_i = \lambda_i e_i : (\Sigma \text{ EVD})$$

$$\downarrow \{\lambda_1, \lambda_2, \lambda_3\} = \{4, 9, 1\}$$

$$\{e_1, e_2, e_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\Sigma^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Sigma^{-1} e_i = \lambda_i^{-1} e_i \quad (\Sigma^{-1} \text{ EVD})$$

$$\{\lambda_1^{-1}, \lambda_2^{-1}, \lambda_3^{-1}\} = \{1/4, 1/9, 1\}$$

same eigenvectors as for Σ

Solution Problem 1.4

$$a) \hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) \quad \text{random part}$$

$$E[\hat{f}(x)] = \frac{1}{nh} \sum_{i=1}^n E\left[K\left(\frac{x-x_i}{h}\right)\right] \quad \text{random part}$$

$$= \frac{1}{nh} n \cdot E\left[K\left(\frac{x-x_i}{h}\right)\right]$$

$$= \frac{1}{h} \int_{-\infty}^{\infty} f(x_i) K\left(\frac{x-x_i}{h}\right) dx_i \quad (\text{def. of } E[\cdot])$$

$$= \frac{1}{h} f(x) * K\left(\frac{x}{h}\right) \quad \text{QED}$$

$$b) \text{MSE}(x) = E[(\hat{f}(x) - f(x))^2] \leftarrow E[X^2] = \mu^2 + E[(X - \mu)^2]$$

$$= \underbrace{\{E[\hat{f}(x)] - f(x)\}^2}_{\mu^2} + E\left[\underbrace{(\hat{f}(x) - f(x)) - (E[\hat{f}(x)] - f(x))}_{\text{random part}}^2\right]$$

$$= \underbrace{\{E[\hat{f}(x)] - f(x)\}^2}_{\mu^2} + E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

$$= \underbrace{\{E[\hat{f}(x)] - f(x)\}^2}_{\text{systematic bias error}} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{random variance}}$$

remember
 $E[f(x)] = f(x)$

big $h \Rightarrow$ big bias + small variance and vice versa

QED